Sam Tang

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Education

University of Waterloo September 2022 – Expected April 2027

Bachelor of Computer Science

- University of Waterloo President's Scholarship of Distinction · Faculty of Mathematics Scholarship

Experience

Cerebras

AI Software Developer Intern January 2026 – Present

- Root-caused and fixed a long-standing wafer stall issue by designing targeted tests and using bisection to isolate the regression, improving system stability for workloads
- Built a weight-corruption test harness to simulate real system failures, enabling broader fault-injection coverage through **MLIR**-based weight buffer calculation, **gRPC/Protobuf** backend integration and bit-level corruption logic
- Brought up a new model in the simulated system test environment compatible with the latest runtime design, reducing test turnaround time by about **80%** by removing queue delays on shared real systems

Huawei

Research Software Developer Intern · Virtual Keyboard Core Development February 2025 – August 2025

- Delivered Matebook Fold virtual keyboard system features at scale (**200k+** users) by owning **C++** development end-to-end from design docs through implementation, experimentation and release
- Broadened multi-finger touchpad support (**2 new gestures**, **9 transitions**) by refactoring the touchpad state machine
- Reduced false taps from **25%** → **5%** and improved gesture success by **27%** through signal filtering and input debouncing
- Enabled on-device automated QA input testing by adding a new input-injection hook into the event-handling pipeline

Research Software Developer Intern · AI/HCI Prototype Development January 2025 – February 2025

- Earned approval to continue development of a novel **AI/HCI** prototype by building a **JavaScript Chrome extension** + **ArkTS** (HarmonyOS) companion app to demonstrate product capabilities
- Improved gesture recognition accuracy from **40%** → **80%** by optimizing $P point-cloud template matching and storage

AutoTrader.ca May 2024 – August 2024

Full-stack Developer Intern

- Shipped a new pre-qualification flow under tight release cycles (consistent closure of high-priority tickets) by implementing features in **Angular** + **.NET** within an agile environment
- Strengthened **API** security boundaries by implementing **server-to-server authentication** from **API controllers** to **Azure APIM** with **Azure B2C JWT** generation
- Cleaned up production data (**7,000+** invalid users removed) by using **Entity Framework** + **Azure AI Search** to accurately identify and delete invalid accounts

Zomp April 2023 – August 2023

Full-stack Developer Intern

- Improved backend scalability (storage bottlenecks removed) by migrating the **.NET** backend to **Azure Cloud**
- Cut **API** response time by **30%** by designing **REST APIs** that stream data efficiently to/from the **Azure** backend
- Reduced an internal workflow from minutes to seconds by building an in-app tool that converts Excel data into an interactive UI component

Technical Skills

Languages: C++ · C# · Python · JavaScript/TypeScript · Java · CSS · HTML · SQL · C

Frameworks: FastAPI · React.js · .NET · Bootstrap · Entity Framework Core · ArkTS · Protobuf · gRPC · Angular

Technologies: Git · MS SQL Server · Azure · MySQL · Pytest · GTest · Redis · Bash